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README



exp-3-digital-image-processing

Histogram Equalization Using OpenCV (Grayscale & Color Images)

Aim

To write a Python program using OpenCV to perform histogram equalization on both grayscale and color images to enhance image contrast and brightness.

The program performs the following operations:

- Read and display a grayscale image
- Plot histogram of the grayscale image
- Apply histogram equalization on grayscale image
- Read and display a color image
- Plot histogram of B, G, R channels
- Convert image to HSV color space
- Apply histogram equalization on the Value (V) channel
- Convert the enhanced image back to BGR format
- Display original and enhanced images with histograms

Software Used

- Anaconda – Python 3.7
- Jupyter Notebook / VS Code
- OpenCV (cv2)
- NumPy
- Matplotlib

Algorithm

Step 1:

Import the required libraries: OpenCV, NumPy, and Matplotlib.

Step 2:

Read the image `parrot.jpg` in grayscale format.

Step 3:

Display the grayscale image and plot its histogram.

Step 4:

Apply histogram equalization using `cv2.equalizeHist()` to enhance contrast.

Step 5:

Display original grayscale image, its histogram, enhanced image, and its histogram using a 2×2 grid.

Step 6:

Read the same image in color format.

Step 7:

Split the image into B, G, R channels and plot their histograms.

Step 8:

Convert the image from BGR to HSV color space.

Step 9:

Apply histogram equalization on the V (Value) channel.

Step 10:

Merge the channels and convert the image back to BGR format.

Step 11:

Display original color image, histogram, enhanced image, and enhanced histogram using a 2×2 grid.

Program

Developed By:

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```
import cv2
import numpy as np
import matplotlib.pyplot as plt

img = cv2.imread('TikTok - Make Your Day.jpg', cv2.IMREAD_GRAYSCALE)

plt.imshow(img, cmap='gray')
plt.title('Original GrayScale Image')
plt.show()

plt.hist(img.ravel(), 256, range=[0,256])
plt.title('Original GrayScale Image Histogram')
plt.show()

img_eq = cv2.equalizeHist(img)

plt.hist(img_eq.ravel(), 256, range=[0,256])
plt.title('Equalized Histogram')
plt.show()

img = cv2.imread('TikTok - Make Your Day.jpg', cv2.IMREAD_COLOR)
img_hsv = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)

img_hsv[:, :, 2] = cv2.equalizeHist(img_hsv[:, :, 2])

plt.subplot(121)
plt.imshow(img[:, :, :-1])
plt.title('Original Color Image')

plt.subplot(122)
plt.imshow(img_eq[:, :, :-1])
plt.title('Equalized Image')

plt.show()

plt.figure(figsize=[12,10])
```



```
plt.subplot(221)
plt.imshow(img[:, :, ::-1])
plt.title('Original Color Image')

plt.subplot(222)
plt.imshow(img_eq[:, :,])
plt.title('Equalized Image')

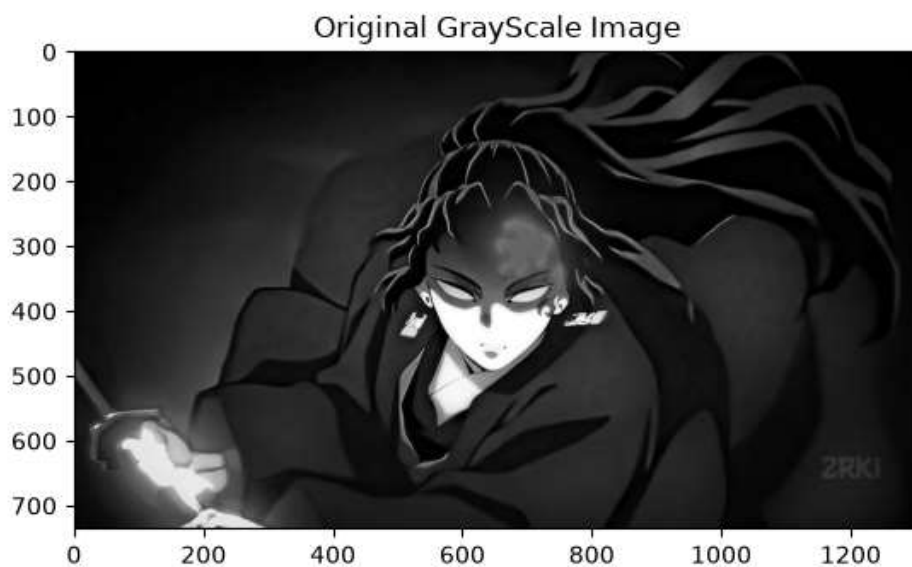
plt.subplot(223)
plt.hist(img.ravel(), 256, range=[0,256])
plt.title('Original Histogram')

plt.subplot(224)
plt.hist(img_eq.ravel(), 256, range=[0,256])
plt.title('Histogram Equalized')

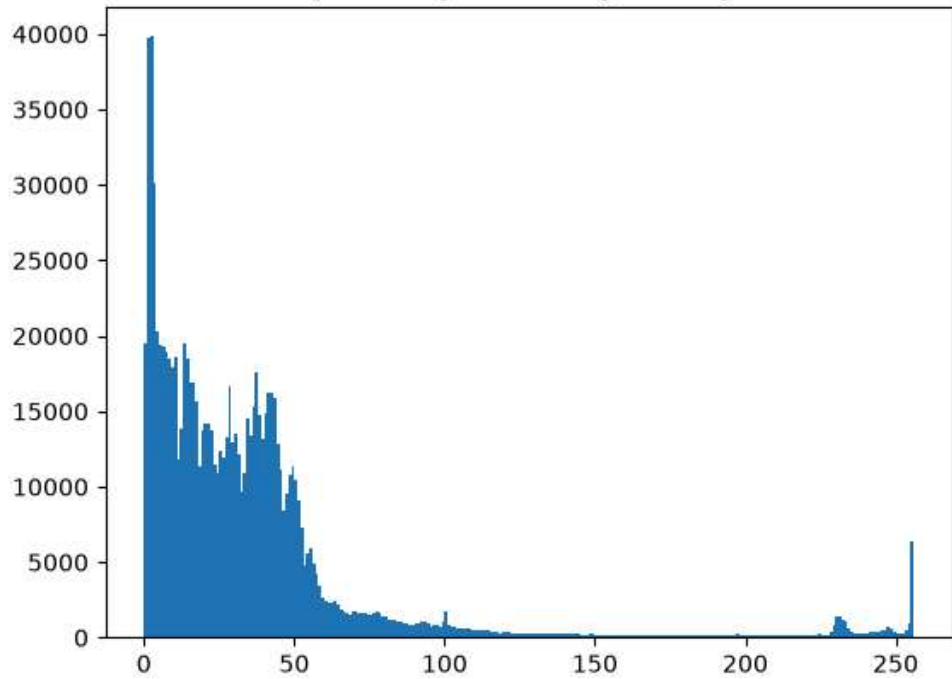
plt.show()
```

Output

original scale img



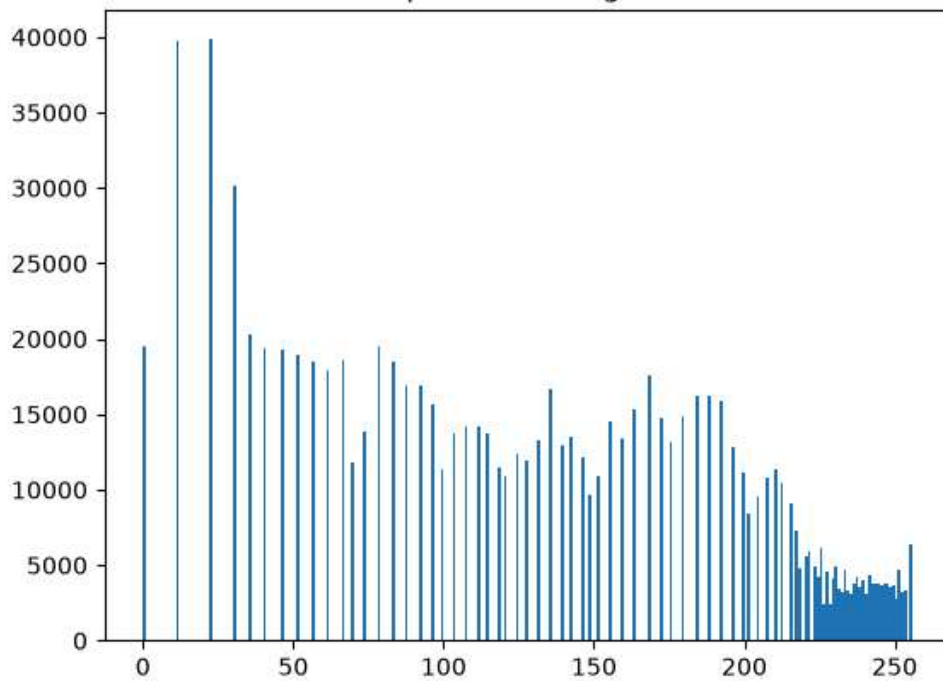
Original GrayScale Image Histogram



original gray scale img

equalized histogram

Equalized Histogram

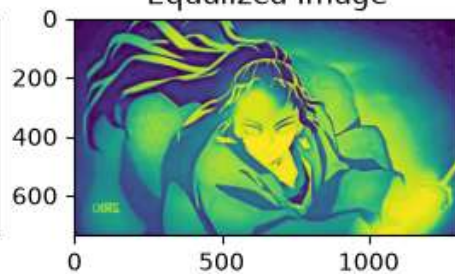


Original Color Image

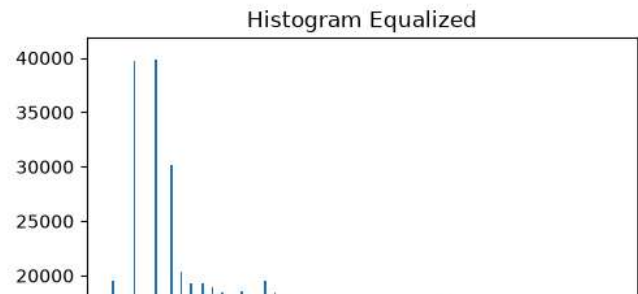
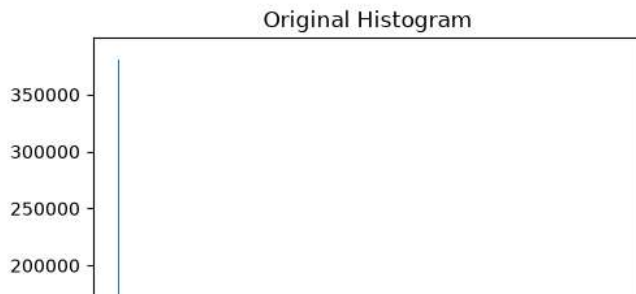
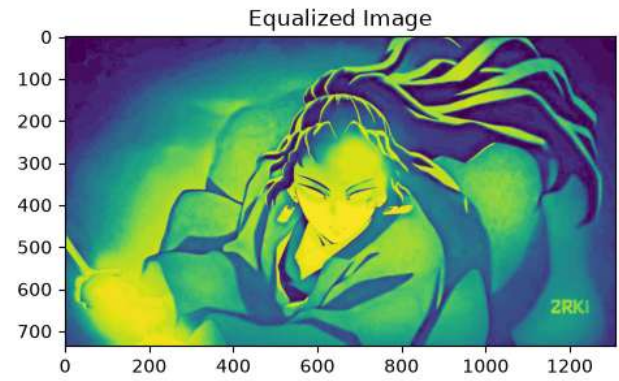


original color img

Equalized Image



orgial color img



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